Problem 4

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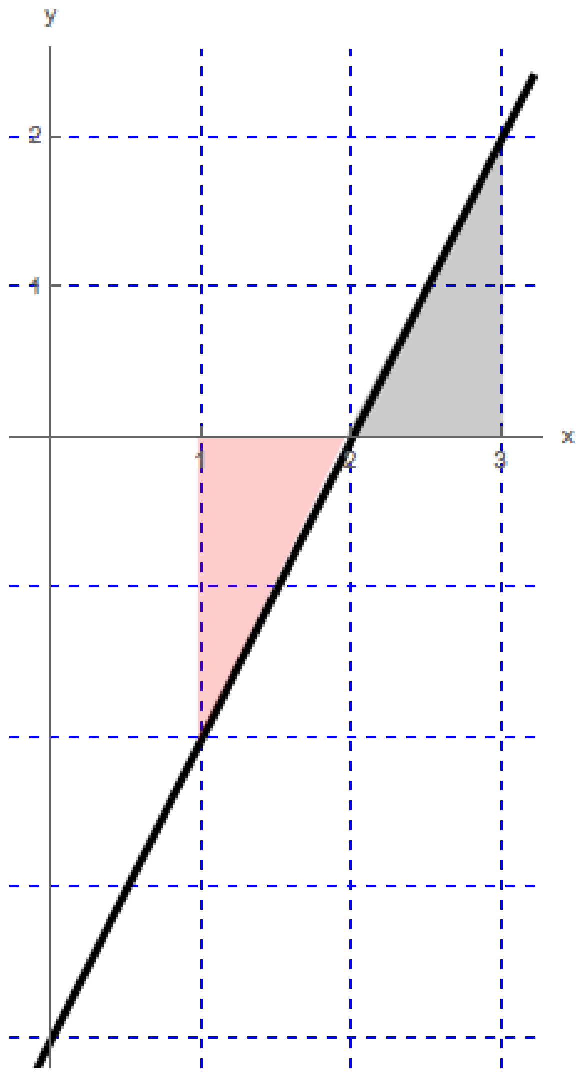
Problem 1 *Use geometry to evaluate the definite integral. Sketch the graph of the function and shade the relevant regions.*

(a) $\int_1^3 (2x - 4)\, dx$



Explanation.

We want to find the net area of the shaded region. We have two identical triangles, each with base 1 and height 2. Since the triangles will have identical area and one lies above and the other below the x-axis, the total area will be 0.

Therefore, $\displaystyle\int_1^3 (2x-4)\,dx = 0$

(b) $\displaystyle\int_1^3 |2x-4|\,dx$

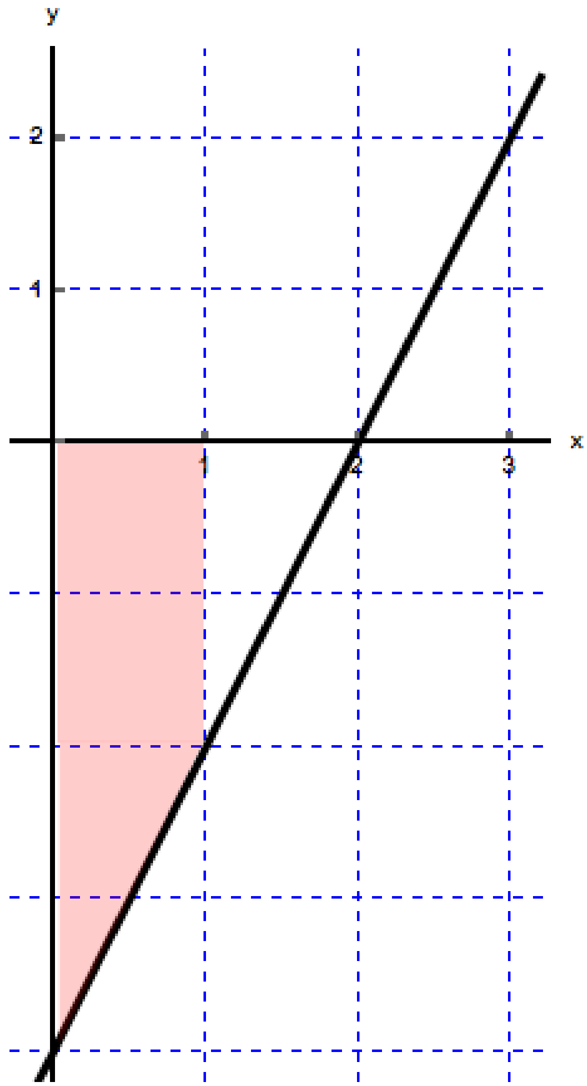
Explanation. The functions changes the sign at $x = 2$. Therefore, we will split the integral into two integrals:

$$\int_1^3 |2x-4|\,dx = \int_1^2 |2x-4|\,dx + \int_2^3 |2x-4|\,dx.$$

Since the function is negative for $x < 2$ and positive for $x > 2$, we write

$$\int_1^3 |2x-4|\,dx = -\int_1^2 (2x-4)\,dx + \int_2^3 (2x-4)\,dx = 1 + 1 = 2.$$

(c) $\displaystyle\int_0^1 (2x-4)\,dx$

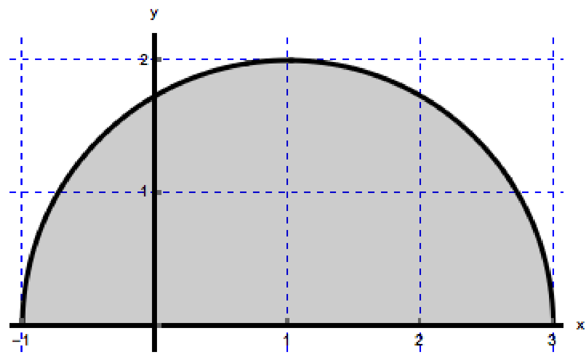


Explanation.

Here we need to find the area of a trapezoid. The trapezoid is made up of a rectangular top with area $(2)(1) = 2$ and triangle bottom with area $(1/2)(1)(2) = 1$. This gives us an area of 3, but, since f is negative on the interval $(0, 1)$, the value of the integral is the negative of this area.

Therefore, $\displaystyle\int_0^1 (2x - 4)\,dx = -3$

(d) $\displaystyle\int_{-1}^{3} \sqrt{4 - (x-1)^2}\,dx$



Explanation.

We are looking for the area of the entire semi-circle above the x-axis. This is a circle with radius 2 so we have $(1/2)\pi 2^2 = 2\pi$

$$\int_{-1}^{3} \sqrt{4-(x-1)^2}\, dx = 2\pi$$